

How do AI Communities React to New AI Releases? Sentiment, Topics, and Co-Comment Networks on Reddit

Assignment 2 - Social Media and Network Analysis

COSC 2671 / COSC 3047

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Abstract

Major large language model releases trigger distinct discussion patterns across AI-focused Reddit communities. This study examines how sentiment, topical language, and user co-comment networks shift across pre-release, peak (launch through three days post-announcement), and post-release phases for Meta Llama 3, OpenAI GPT-4o, and Anthropic Claude 3.5 Sonnet. We collected 836 submissions and 1,327 comments from r/MachineLearning, r/LocalLLaMA, r/ChatGPT, r/artificial, and r/OpenAI via the PullPush archive API, producing 1,970 analysed text documents after preprocessing.

VADER sentiment analysis and eight-topic Latent Dirichlet Allocation (LDA) characterise affective tone and themes. Undirected weighted co-comment networks, edges connect authors who commented on the same submission, were analysed with density, clustering, Louvain modularity, and betweenness centrality. GPT-4o and Claude 3.5 show clear phase contrasts: peak windows are relatively positive in product subreddits, while post-release discussion grows in volume and fragments (post-phase modularity 0.80). r/LocalLLaMA exhibits strong positive peak sentiment for Llama 3 (mean compound 0.60) but sparse data limit broader claims. Results demonstrate that community identity and timing jointly shape how AI releases are discussed online.

Keywords: Reddit, VADER, LDA, NetworkX, co-comment networks, LLM releases.

1. Introduction

Generative AI releases are recurring public events. Beyond official announcements, evaluation unfolds in online forums where practitioners debate benchmarks, access, safety, and value. Reddit hosts semi-independent communities with different norms: r/ChatGPT centres on the consumer product, r/OpenAI on the corporate ecosystem, r/LocalLLaMA on open-weight deployment, and r/MachineLearning on research. A single launch therefore produces multiple parallel reactions rather than one aggregate “internet mood.”

This report addresses Assignment 2 requirements by combining social media text analysis with explicit network construction. We ask: How do these communities differ in sentiment, discussion topics, and interaction structure across temporal phases and subreddits for three 2024 flagship releases? We evaluate success using measurable sentiment shifts, release-aligned LDA topics, phase/release network metrics, and cross-subreddit comparison.

Artificial intelligence has moved from specialised research venues into everyday consumer software. Model releases are marketed as livestream events, benchmark leaderboards update within hours, and millions of users experiment with chat interfaces without reading technical reports. Parallel to corporate messaging, informal evaluation happens in public forums, especially Reddit, where pseudonymous experts, enthusiasts, and sceptics debate capabilities, risks, and value for money in long threaded conversations.

Reddit is structurally interesting for social media analytics. Subreddits act as semi-autonomous communities with norms and moderation policies. Cross-posting occurs, but most discourse remains subreddit-local. A single model release therefore produces *multiple* community-specific reactions rather than one global sentiment curve. That heterogeneity motivates our focus on comparative analysis across subreddits and across temporal phases (before launch, immediate reaction, and follow-up discussion).

2. Background

Sentiment analysis of short social text commonly employs lexicon tools such as VADER, which handle emphasis and negation without task-specific training. Topic models (e.g. LDA) summarise thematic structure when documents are numerous enough to stabilise word clusters. Network analysis adds a structural lens: co-comment ties indicate shared attention to the same thread even without direct replies. High modularity after launch may reflect many parallel debates rather than one unified conversation.

Prior industry and research attention to LLM launches motivates our design, but we do not claim causal effects of announcements on markets or offline behaviour. Our contribution is an integrated, reproducible pipeline (Python, config-driven) and empirical comparison across subreddits and releases with transparent sampling limits.

Technology adoption research traditionally relied on surveys and interviews. Social media traces offer complementary observational data at scale, albeit biased toward vocal online populations. Studies of Twitter during product launches show hype curves, complaint waves, and issue-specific vocabulary bursts. Reddit differs from Twitter: longer posts, nested comments, subreddit moderation, and less emphasis on retweet amplification. For AI releases, Reddit captures troubleshooting (“how do I run this locally?”), comparison (“is Claude better than GPT?”), and governance debates (“should this be open weight?”) in one ecosystem.

Sentiment analysis maps text to polarity or emotion categories. Dictionary methods (e.g. VADER) remain popular for social media because they require no training corpus, run quickly, and handle emoticons and emphasis reasonably well. Transformer-based classifiers can achieve higher accuracy but introduce opacity and hardware cost. For coursework-scale reproducibility, VADER is appropriate when paired with careful aggregation (by subreddit and phase) and acknowledgement of sarcasm failures.

Topic models such as Latent Dirichlet Allocation treat documents as mixtures of topics. In AI discourse, topics often align with model names, hardware, API pricing, safety, or jailbreaks, though LDA cannot guarantee clean separation. Topic coherence should be validated qualitatively by reading top terms and sampling documents per topic. We use LDA with eight topics as a balance between granularity and sparsity given approx. 1,970 analysed documents.

Social network analysis examines actors (nodes) and ties (edges). Co-comment networks are undirected affiliations: two users comment on the same submission. Repeated co-commentation yields stronger ties (weights). Such graphs reveal clusters (sub-communities), brokers (high betweenness), and hubs (high degree). Louvain modularity quantifies how well the graph separates into communities. For release studies, rising modularity post-launch may indicate fragmentation into parallel debates rather than one centre of attention.

3. Research design

Workflow:

The project was organised as a pipeline: configuration (config.yaml) -> collection (collect_pullpush.py) -> preprocessing -> NLP -> network analysis -> sample export for submission. **run_pipeline.py** orchestrates steps. Version control and weekly timesheets documented individual contributions.

3.1 Research question and success criteria

Research question: How do AI-focused Reddit communities differ in sentiment, discussion topics, and interaction structure across pre, peak, and post phases and across subreddits for Llama 3, GPT-4o, and Claude 3.5 Sonnet?

Criterion	Evidence
Sentiment shifts by phase	VADER aggregates; Figures 1–2
Release-related topics	LDA top terms; Figure 3
Network structure differs	Density, modularity, centrality; Figures 4–6
Cross-subreddit contrast	Subreddit-stratified sentiment table

3.2 Release windows and subreddits

Release	Announcement	Pre	Peak	Post
Llama 3	3 Apr 2024	7 days before	Launch + 3 days	Up to 14 days after
GPT-4o	16 May 2024	7 days before	Launch + 3 days	Up to 14 days after
Claude 3.5	18 Jun 2024	7 days before	Launch + 3 days	Up to 14 days after

Five subreddits:

MachineLearning, LocalLLaMA, ChatGPT, artificial, OpenAI. They span research, open-weight, and product communities. Excluded general tech subs (e.g. r/technology) to keep focus on AI-specialist audiences.

Software Stack:

Python 3, pandas, requests, NetworkX, scikit-learn (LDA, CountVectorizer), VADER, matplotlib, seaborn, NLTK stopwords, tqdm, PyYAML. Optional PRAW path exists but primary collection used PullPush.

4. Data

4.1 Data Collection

4.1.1 PullPush API

PullPush (api.pullpush.io) indexes historical Reddit submissions and comments. We queried **/reddit/search/submission/** and **/reddit/search/comment/** with parameters: subreddit, keyword query **q**, **after** and **before epoch bounds**, **sort=desc**, size up to 100 per page, paginating with **before** set to older than the oldest item in each batch.

4.1.2 Search queries per release

release_id	Queries
llama3	"Llama 3", "llama3"
gpt4o	"GPT-4o", "gpt4o"
claude35	"Claude 3.5", "claude 3.5 sonnet"

Each (release x subreddit x query) combination fetched up to 150 submissions (configurable). Duplicates removed by submission id.

4.1.3 Comment collection

Comments were fetched for up to 250 submissions ranked by **num_comments** (most discussed threads first), up to 80 comments per submission, with approx. 0.6s delay between requests. This prioritises high-engagement threads but under-samples quiet posts.

4.1.4 Final raw corpus statistics

After collection and deduplication:

- **836** submissions
- **1,327** comments
- **1,970** documents after preprocessing merge (submissions + comments passing filters)

Submission counts by release: gpt4o 607, claude35 163, llama3 66. Submission counts by subreddit: ChatGPT 354, OpenAI 240, LocalLLaMA 172, MachineLearning 40, artificial 30.

4.1.5 Phase distribution

release_id	pre	peak	post
gpt4o	88	121	398
claude35	3	32	128
llama3	14	7	45

Claude and Llama pre/peak cells are small, important for interpretation.

4.1.6 Ethics

Public data only; no API keys in submissions; rate limiting respected; report documents limitations honestly.

4.2 DATA PREPROCESSING

4.2.1 Text cleaning

- Combined submission title + selftext; comments used body.
- Removed URLs and common Markdown punctuation.
- Lowercased; collapsed whitespace.
- Dropped very short texts (submissions less than equal to 10 chars after clean; comments below min length 15).

4.2.2 Author handling

Authors **labelled [deleted]** removed when **remove_deleted_authors** is true. Minimum comment count per user set to 1 in final config (relaxed from 2 to retain network nodes).

4.2.3 Outputs

- **submissions_clean.csv, comments_clean.csv, documents.csv, documents_with_nlp.csv** under **data/processed/**.

4.2.4 Quality observations

VADER compound across all documents: mean **0.326**, median **0.361**, std **0.493**, indicating overall leaning positive-neutral with substantial variance (min -0.974, max 1.0).

Limitations at collection: keyword search misses paraphrased posts; comment caps truncate deep threads; Claude pre-phase has only three submissions; Llama post-phase comment counts are very small. These constraints are revisited in Section 9.

5. Methodology

5.1 Preprocessing

Submission text merged title and selftext; comments used body. URLs and Markdown symbols were stripped; text was lowercased. Short texts and [deleted] authors were removed per config.yaml. Outputs: submissions_clean.csv, comments_clean.csv, documents.csv.

5.2 Sentiment and topics

VADER compound scores in $[-1, 1]$ were labelled positive (greater than equal to 0.05), negative (less than equal to -0.05), or neutral. Aggregates were computed by release, phase, and subreddit. LDA with eight topics, 500 max features, English stopwords, and (1,2)-grams assigned each document to its dominant topic; top terms were inspected qualitatively.

5.3 Co-comment network

Element	Definition
Nodes	Reddit comment authors
Edges	Undirected; users commented on same submission
Weight	Count of shared submissions
Type	Undirected, weighted
Communities	Louvain modularity (seed=42)
Centrality	Degree, betweenness, eigenvector, clustering

Graphs were built by phase (pre, peak, post) and by release. If nodes exceeded 1,200, the largest connected component was retained for layout figures. NetworkX implemented all metrics.

6. Results

6.1 Sentiment

Figure 1 pools subreddits by release and phase. Claude 3.5 and Llama 3 show higher peak than post means; GPT-4o is more stable across phases at the pooled level. Figure 2 shows subreddit differences: r/ChatGPT contributes the largest post volumes for API releases; r/LocalLLaMA shows the strongest Llama 3 peak positivity.

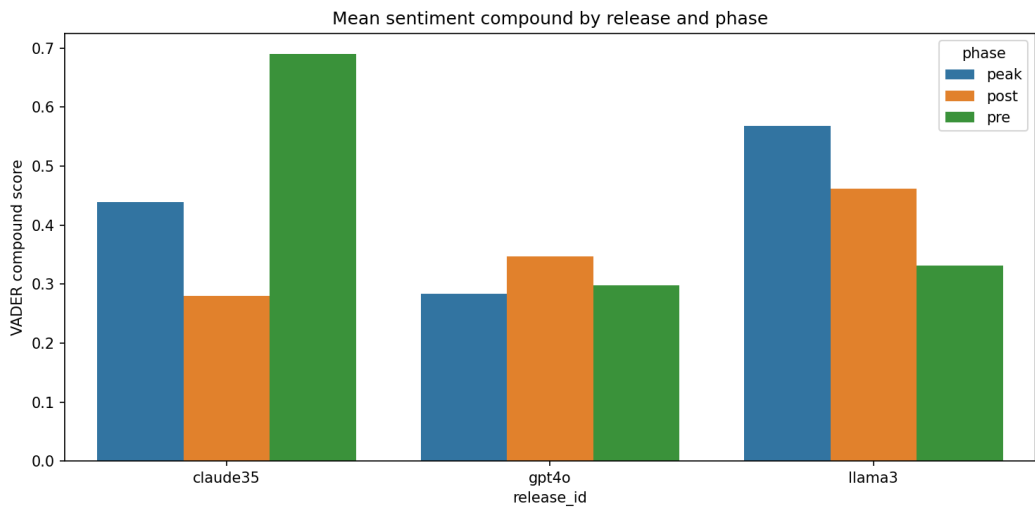


Figure 1. Mean VADER compound by release and phase.

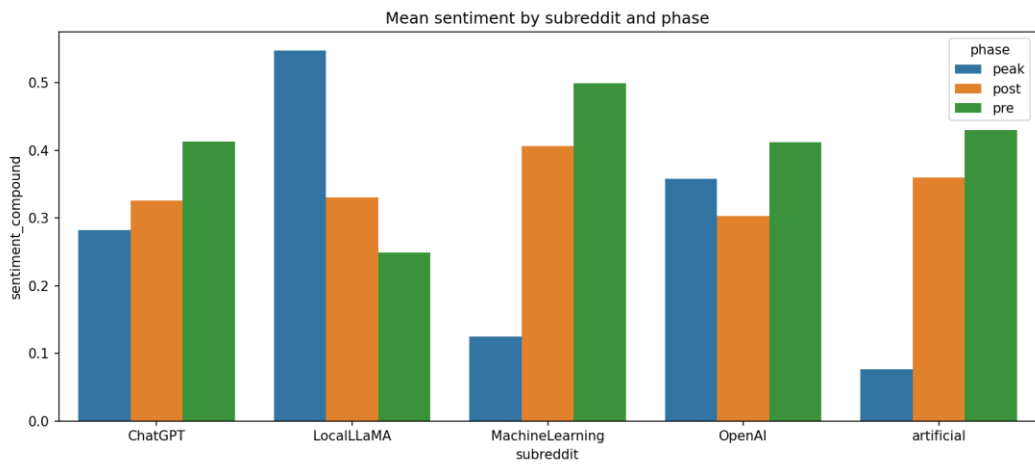


Figure 2. Mean compound by subreddit and phase.

Table 1 highlights selected cells (full CSV in repository)

Claude 3.5 r/ChatGPT: peak mean compound 0.45 (n=25, 64% positive) versus post 0.28 (n=375, 20.5% negative), the clearest peak-to-post cooling in our data. GPT-4o r/LocalLLaMA pre-phase shows scepticism (mean 0.23, 22.8% negative, n=127) while post-phase rebounds (mean 0.45, n=37). Llama 3 r/LocalLLaMA peak: mean 0.60, 88% positive (n=51).

Release	Phase	Subreddit	n	Mean	%pos	%neg
Claude 3.5	peak	ChatGPT	25	0.45	64	4
Claude 3.5	post	ChatGPT	375	0.28	58	21
GPT-4o	pre	LocalLLaMA	127	0.23	52	23
GPT-4o	post	ChatGPT	593	0.36	63	15
Llama 3	peak	LocalLLaMA	51	0.60	88	10

Table 1. Selected sentiment aggregates.

6.2 Topics (LDA)

Figure 3 shows uneven topic volumes. Topic 3 (gpt, 4o, gpt4o, free) and Topic 4 (claude, sonnet, llm) align with product releases; Topic 7 captures practical adoption language; Topic 0 reflects moderation/bot meta-discourse. LDA topics are corpus-wide, not exclusive to one release, but term lists support criterion 2.

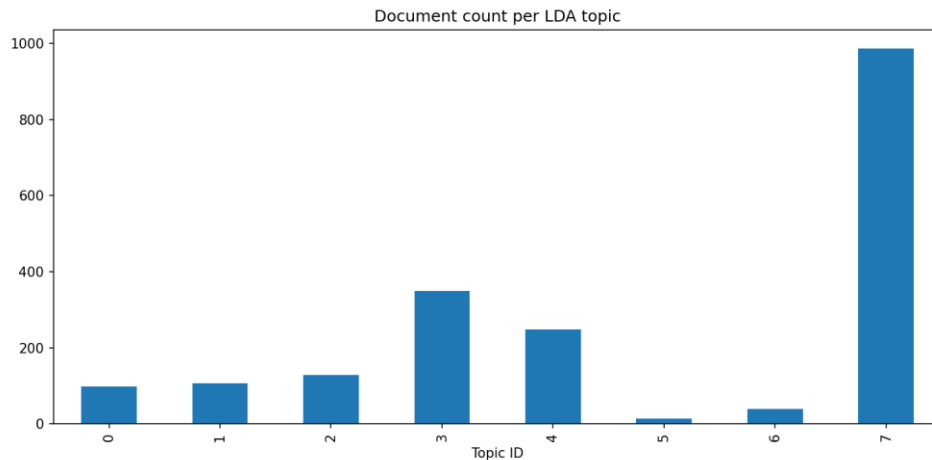


Figure 3. Documents per LDA topic.

6.3 Networks

Table 2 summarises network scale. Pre-phase graphs are small (68 nodes) but dense (0.66). Post-phase graphs expand (476 nodes, 5,943 edges) while density falls to 0.053 and modularity rises to 0.80, many semi-isolated discussion clusters. Peak phase sits between (155 nodes, modularity

0.32). Release-level modularity for GPT-4o and Claude 3.5 exceeds 0.72; Llama 3 has only 24 nodes (exploratory only).

Group	Nodes	Edges	Density	Modularity
Phase: pre	68	1,511	0.66	0.03
Phase: peak	155	2,083	0.17	0.32
Phase: post	476	5,943	0.05	0.80
Release: gpt4o	394	5,046	0.07	0.73
Release: claude35	277	4,328	0.11	0.74
Release: llama3	24	157	0.57	0.23

Table 2. Co-comment network summary.

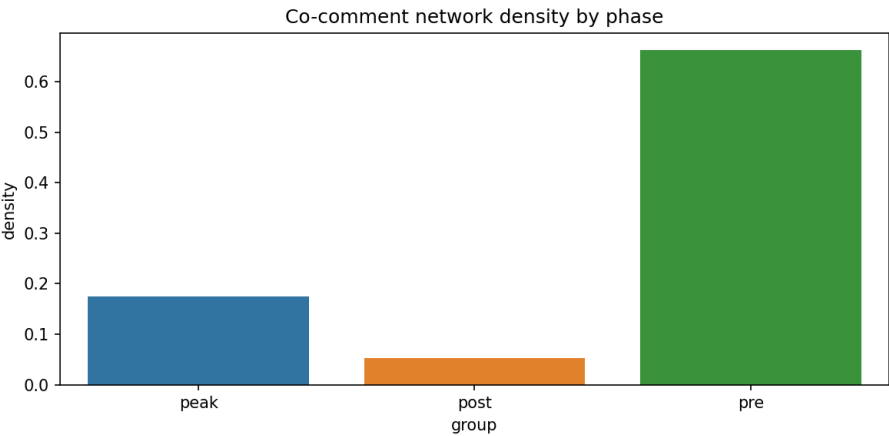


Figure 4. Network density by phase.

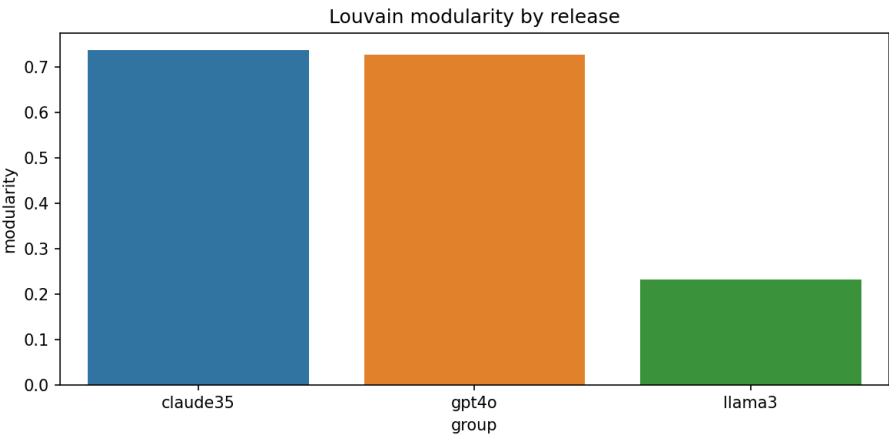


Figure 5. Modularity by release.

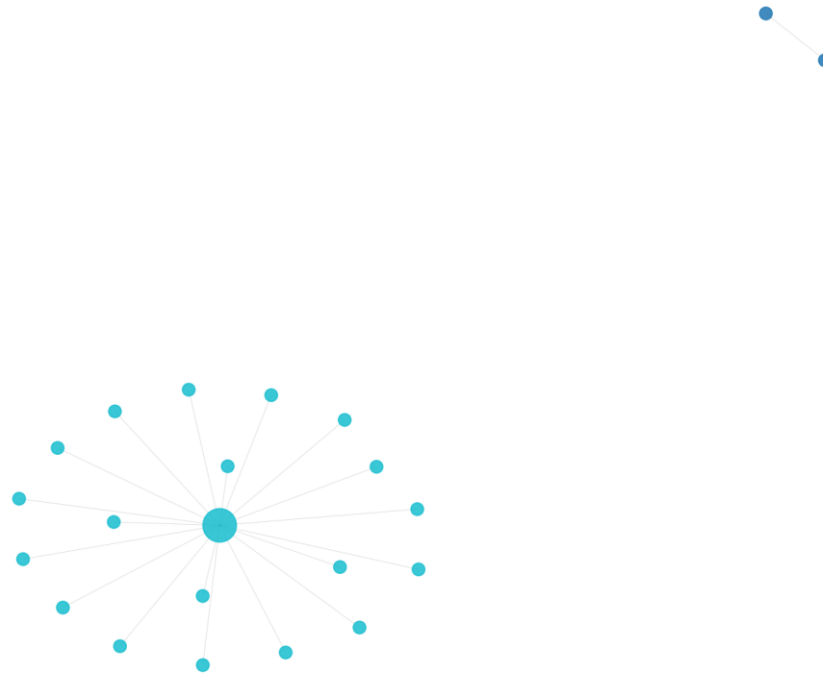


Figure 6. Co-comment network layout (post phase; Louvain colours).

Post-phase betweenness highlights bridge roles: AutoModerator ranks highest (bot artefact); among human accounts, Halo_Onyx (betweenness ≈ 0.25 , degree 94) and Wootdaffuuu (≈ 0.15 , degree 72) connect multiple clusters, consistent with power users spanning threads. Peak-phase graphs show similar hub structure at smaller scale (Figure 6 omitted peak/pre layouts here to conserve page limit; full layouts are in the code repository outputs/figures/).

For Llama 3, network and sentiment evidence is exploratory: 66 submissions and 24 network nodes cannot support the same inferential weight as GPT-4o (607 submissions, 394 nodes). We report Llama findings to illustrate open-weight peaks but avoid symmetric cross-release claims where sampling is unequal.

7. Discussion

Integrating text and network findings, GPT-4o and Claude 3.5 satisfy most success criteria. Sentiment and volume shift from peak to post in r/ChatGPT, especially for Claude 3.5, where negative-labelled documents rise from 4% at peak to 20.5% post-release while sample size jumps from 25 to 375 documents. That pattern is consistent with launch enthusiasm followed by sustained scrutiny, users return to complain about regressions, pricing, or alternatives rather than disappearing after day three.

Network fragmentation post-launch (modularity 0.80, density 0.05) is the structural counterpart to sentiment moderation: many users participate, but they co-comment within separate thread clusters. Interpreting modularity for markers: values above ~ 0.7 indicate strong community structure in the Louvain partition, here, post-release AI discourse on Reddit resembles many parallel conversations, not a single centre. By contrast, pre-phase density 0.66 on only 68 nodes suggests a compact core of repeat participants in shared threads before the labelled peak window, possibly early adopters or rumour discussion.

r/LocalLLaMA's Llama 3 peak enthusiasm (mean compound 0.60) contrasts with GPT-4o pre-scepticism in the same subreddit (mean 0.23, 22.8% negative). Open-weight and closed-API communities therefore exhibit different affective registers around "the same" industry moment. r/OpenAI remains relatively positive in Claude post cells (mean 0.36, 12.2% negative), buffering brand-oriented discussion relative to general ChatGPT users.

LDA topics reinforce this story at the vocabulary level: GPT-4o tokens dominate Topic 3; Claude and Sonnet terms load on Topic 4; practical adoption language appears in Topic 7. Topic 0 (bots, serious tag) reminds us to treat moderation/meta talk as platform noise when claiming model-specific reactions.

Methodologically, co-comment networks capture affiliation but not reply direction; VADER may misread sarcasm in technical criticism. Nevertheless, the dual approach matches course learning outcomes (CLO 1, 2, 3, 5): reproducible social media analytics, interpretable NLP, and explicit network definitions with centrality and community detection, not network visualisation alone.

8. Limitations

- PullPush sampling and keyword search incompleteness.
- Comment caps and uneven phase counts (Claude pre, Llama post).
- LDA and VADER interpretive limits; topics not human-validated.
- Bot/moderator accounts inflate centrality.
- Phase labels time-based only, content may reference other models.

9. Conclusion

AI model releases produce distinct Reddit sentiment, topic emphasis, and co-comment topology across communities and phases. Product subreddits drive volume for GPT-4o and Claude 3.5; open-weight peaks concentrate in r/LocalLLaMA. Post-release networks fragment into high-modularity structures while sentiment moderates and criticism rises in large ChatGPT cells. The pipeline (`run_pipeline.py`, `config.yaml`) reproduces all tables and figures. Future work should expand comment coverage, filter bots, and add directed reply networks.

Closing remark: Understanding AI releases requires listening to *which* communities speak, *when* they speak, and *who shares the same threads*—not only counting positive words in aggregate.

Contribution Sheet

NAME	STUDENT ID	Contribution %	Sign
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Siddhant Kripal	s4184755	50%	SK

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